

Business Case — Fort Frances & Area Wireless Broadband Network

Prepared for: Jam21.net (in coordination with Rainy River Futures Development Corporation) **Prepared by:** Northern Wireless Inc. (NWI) **Date:** June 19, 2026 **Reference:** NWI-BC-FF-2026 · Rev 0

1. Purpose

This business case supports Jam21.net's decision to build a fixed-wireless broadband network serving Fort Frances and the surrounding area. It summarizes the recommended solution, the capital cost to build it, and the operating model that sustains it, so that Jam21.net can proceed to a supply-and-install contract with NWI.

2. Opportunity

Fort Frances and the neighbouring communities have limited affordable broadband. A fixed-wireless network lets Jam21.net deliver high-speed internet to schools, businesses and residents at a fraction of the cost of wired build-out, with fast deployment and the ability to grow capacity as demand grows.

3. Recommended Solution

A complete turnkey network supplied, installed and commissioned by NWI:

- **Backbone:** 5.8 GHz point-to-point / point-to-multipoint links from a central tower.
- **Last mile:** 802.11-based customer premise equipment (CPE) at each subscriber.
- **Capacity:** designed for **60 Mbps**, against an estimated 5-year area demand of **50 Mbps** — headroom to extend service to nearby towns.
- **Coverage:** up to **8 break-out locations** feeding local last-mile cells.
- **Core services:** bandwidth-control, primary/secondary servers and T1 hand-off, ready to integrate with Jam21.net's existing ISP (RADIUS, email, web).

Jam21.net owns and operates all equipment on completion. NWI trains Jam21.net staff to run and maintain the network.

4. Capital Investment (Supply & Install)

One-time turnkey cost to build the network, covering 100 initial subscriber connections:

Work package	Scope	Amount (CAD)
Engineering & Admin	Site study, design, drawings, technical manual, procurement	\$10,480
Tower	Supply, installation, power & lighting	\$17,000
Tower POP (head-end)	Antennas, amplifiers, AP enclosure, cabling, install	\$5,500
Wireless Backhaul	Backhaul radios, cabling, install both ends	\$5,500

Work package	Scope	Amount (CAD)
Core Wired Network	Bandwidth control, switch, primary/secondary servers, T1/router	\$19,500
Premise Equipment	100 subscriber radios, antennas, cable, connectors	\$31,200
Commissioning & Testing	On-the-ground and installed on-site testing	\$8,320
Total supply & install		\$97,500

Pricing is firm and itemized; the full line-item breakdown carries into the supply contract. Government rural-broadband subsidies, where secured, offset part of this capital cost.

5. Operating Model (Monthly)

The network is designed to **break even at launch** on a modest anchor base, then generate surplus as subscribers are added.

Monthly operating costs

Item	Monthly (CAD)
T1 backhaul (internet transit)	\$2,700
Network operations (part-time, ~4 hrs/day @ \$15/hr)	\$1,600
Total	\$4,300

Monthly recurring revenue at launch

Segment	Accounts	\$/mo each	Monthly (CAD)
Schools (256 kbps)	2	\$300	\$600
Business	10	\$100	\$1,000
Residential	20	\$45	\$900
Anchor tenant — Wasaya	1	\$800	\$800
Anchor tenant — Mishkeegogamang	1	\$1,000	\$1,000
Total			\$4,300

Net at launch: \$0/month (break-even). Fixed costs (T1 + part-time ops) do not rise materially with new subscribers, so most revenue beyond the launch base is surplus.

6. Growth & Payback

Each additional subscriber is largely margin once the fixed base is covered:

- **+40 residential** (@ \$45) and **+10 business** (@ \$100) in Year 2 $\approx$ **+\$2,800/month** (~\$33,600/year) surplus.
- At that growth, operating surplus repays the \$97,500 capital investment in roughly **3 years** on subscriber revenue alone — sooner if broadband subsidies offset capital.

The model scales: the 60 Mbps design and 8 break-out sites allow expansion into adjacent communities without re-engineering the core.

7. Funding

- **Capital:** Jam21.net, coordinated financially through Rainy River Futures Development Corporation, with government rural-broadband programs as available.
- **Operating:** self-funding from subscriber revenue from launch.
- **Tower rental** and **bandwidth resale** remain Jam21.net's responsibility; NWI supplies, installs, tests and hands over the network.

8. Key Risks & Mitigation

Risk	Mitigation
Backbone link outage affects multiple cells	On-hand spares and certified climbers; proven 5.8 GHz radios
Subscriber uptake slower than plan	Break-even at a small anchor base; low fixed cost to hold
Capacity growth outpaces design	60 Mbps design vs. 50 Mbps 5-year demand; bandwidth control per user
Schedule slippage across parties	Single master schedule with milestone dates, agreed before execution

9. Recommendation & Next Steps

The solution is technically sound, break-even from launch, and scalable. **NWI recommends Jam21.net proceed.**

1. Approve this business case.
2. Review NWI's detailed **Proposal** (to be issued **July 3, 2026**).
3. Execute the **Supply & Install Contract** with NWI to begin engineering and procurement.

Prepared by Northern Wireless Inc. — turnkey wireless internet solutions. Figures are illustrative for planning purposes.